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Benchmark Radar Targets for the Validation of Computational Electromagnetics Programs

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Forward

[This issue's contribution is again on benchmarking. The discussion on CARLOS, which I promised you in the previous column, will be delayed until the next issue. Alex Woo, Helen Wang, Michael Schuh, and Michael Sanders have again put together an nice set of measurements. The measurements described refer to three-dimensional perfectly-conducting smooth targets (except for the cone-sphere with the gap). For most targets, only radar-cross-section patterns at two frequencies are shown, but it is my understanding that patterns at other frequencies may be available to qualified readers. Measurements were also collected after application of dielectric coatings on some of these targets. The measurements of the coated targets are not, however, included in this contribution.

I want to thank the authors for providing us the additional data, which I hope that many readers will find to be of interest.]

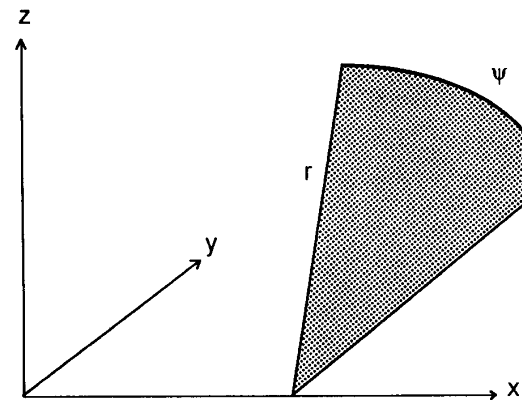

Figure 1. The coordinate system.

Summary

This is the second in a series of articles on Computational Electromagnetics (CEM) validation measurements for the Electromagnetic Code Consortium (EMCC) [1, 2]. This article discusses both the low- and high-frequency measurements of the NASA almond and several other bodies of revolution (BOR), an ogive, a double ogive, a cone-sphere, and a cone-sphere with a gap. Except for the Almond, these are generic simple shapes [3, 4].

Five differently-shaped targets were designed, manufactured, and measured: the NASA almond, ogive, double ogive, cone-sphere and cone-sphere with gap. These were measured from 700 MHz to 16 GHz. The metallic targets are made of aluminum, and were cut by a numerically-controlled mill to maintain the surface precision. Except for the almond target, all the other targets were made in two parts, and joined by sleeves and screws. Please refer to the Naval Warfare Center data reports for more information [5, 6, 7, 8, 9] on the measurement set up, and detailed descriptions of the data.

In the descriptions of the targets in this section, the cylindrical coordinate system shown in Figure 1 is used. The x-axis is along the long dimension, and the z-axis is along the shortest dimension. The polar angle ψ, in the yz-plane, is measured from the xy-plane in the plane perpendicular to the x-axis.

The computer codes used for the comparisons are FERM, a 3-D MoM code [10], and CICERO, a BOR [body of revolution] MoM code [11, 12].

In the figures below, the three dimensional views of the target are faceted for illustration purposes only. These targets are smooth, with C^1 continuity.

Metallic almond

The NASA almond was defined in [13] and [14]. The mathematical description used for this target is as follows:

for $-0.41667 < t < 0$ and $-\pi < \psi < \pi$

$x = dt$ inches

$$y = 0.193333d\sqrt{1-\left(\frac{t}{0.416667}\right)^2}\cos\psi$$

$$z = 0.064444d\sqrt{1-\left(\frac{t}{0.416667}\right)^2}\sin\psi$$

for $0 < t < 0.58333$ and $-\pi < \psi < \pi$

$x = dt$ inches

$$y = 4.83345d\left[\sqrt{1-\left(\frac{t}{2.08335}\right)^2}-0.96\right]\cos\psi$$

$$z = 1.61115d\left[\sqrt{1-\left(\frac{t}{2.08335}\right)^2}-0.96\right]\sin\psi$$

where d = 9.936 inches. The total length of the almond is 9.936 inches. The metallic almond target is shown in Figure 2.

In Figure 3, the RCS values for both horizontal (HH) and vertical (VV) polarizations are plotted in dBSM [dB with respect to one square meter] as a function of the azimuthal angle. The elevation angle is zero, and the almond is measured with its broad side flat. Zero degrees azimuth corresponds to incidence on the tip. We chose to plot the RCS pattern at 1.19 GHz in Figure 3 because, at that frequency, the almond is approximately one wavelength long.

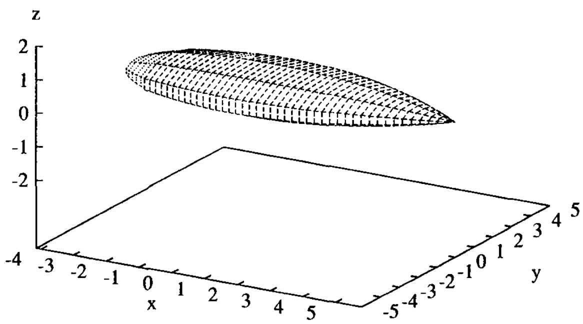

Figure 2. The NASA metallic almond.

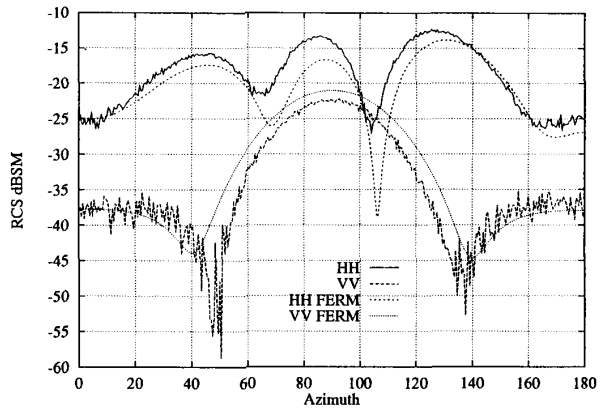

Figure 3. The radar cross section of the NASA Almond at 1.19 GHz. The measured results are labeled "HH" and "VV" for horizontal and vertical polarization, respectively, while "HH FERM" and "VV FERM" denote the computed results obtained using FERM.

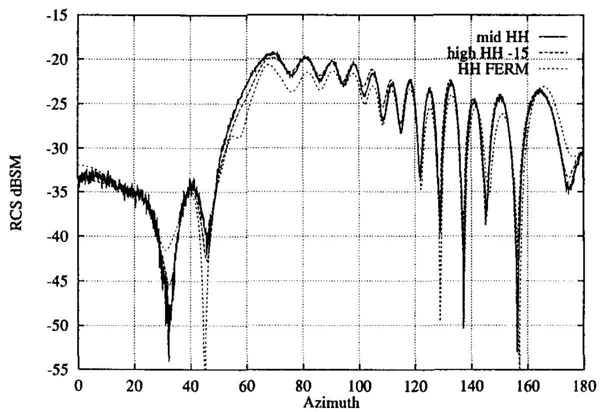

Figure 4. The 9.936 inch NASA almond at 7 GHz, for horizontal polarization: "mid-HH" and "high-HH" denote the measured cases (see the text), and "HH FERM" denotes the computed results.

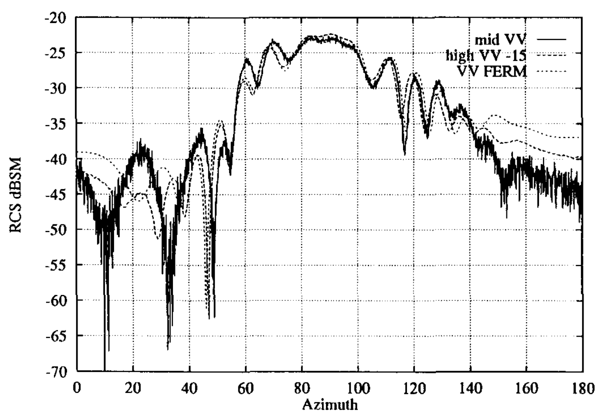

Figure 5. The 9.936 inch NASA almond at 7 GHz, for vertical polarization. The curves are labeled the same as in Figure 4.

The patterns corresponding to frequencies of 7 GHz and 9.92 GHz are given in Figures 4, 5, and 6. The two sets of data shown in Figures 4 and 5, labeled "mid" and "high," refer to the frequency ranges from which the 7 GHz data were taken. The "mid" range corresponds to 2 to 8 GHz, and the "high" range corresponds to 4 to 16 GHz. The almond at 9.92 GHz was specified as a large test case for 3D Method of Moments codes on a workstation. Approximately six unknowns per wavelength were used in Figure 6, and this may explain the differences between measurement and FERM results at 9.92 GHz.

Metallic ogive

The ogival body is a classical RCS test case. High-frequency solutions are available in [3, 4]. The metallic ogive target has a half-angle of 22.62 degrees, a half-length of 5 inches, and a maximum radius of 1 inch, as shown in Figure 7. The analytical expression for the ogive is as follows:

for -5 in $< x <$ 5 in and $-\pi < \psi < \pi$, define

$$f(x)=\left\{\sqrt{1-\left(\frac{x}{5}\right)^2\sin^2(22.62°)}-\cos(22.62°)\right\},\text{ then}$$

$$y=\frac{f(x)\cos\psi}{1-\cos(22.62°)}$$

$$z=\frac{f(x)\sin\psi}{1-\cos(22.62°)}$$

In Figure 8, the RCS for both horizontal and vertical polarization is plotted in dBSM as a function of the azimuthal angle. The elevation angle is zero, and the ogive is positioned as in Figure 7. Zero degrees azimuth corresponds to incidence normal to a tip. Again, the frequency of 1.18 GHz was chosen because the ogive is approximately one wavelength long at that frequency. We note that the horizontally- and vertically-polarized RCS should be equal at 0° and 180° azimuth in Figure 8.

The RCS patterns corresponding to a frequency of 9 GHz are given in Figure 9.

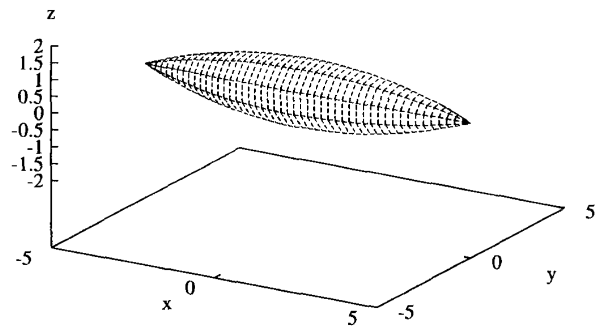

Figure 7. The single ogive.

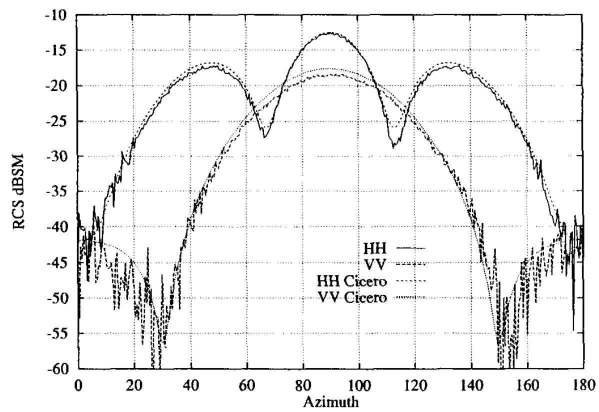

Figure 8. The results for the metallic ogive at 1.18 GHz. "HH" and "VV" denote the measurements for horizontal and vertical polarization, respectively, while "HH Cicero" and "VV Cicero" denote the respective computed results.

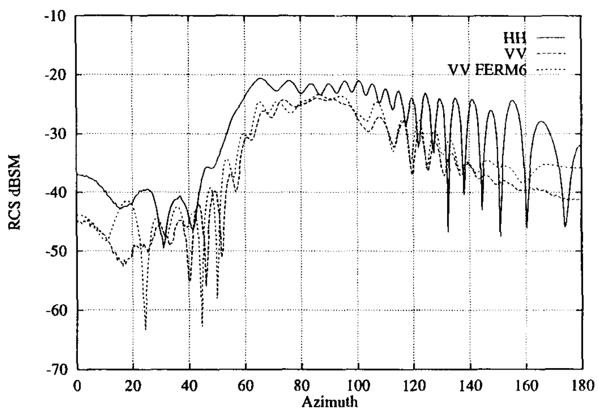

Figure 6. The 9.936 inch NASA almond at 9.92 GHz. "HH" and "VV" denote the measurements for horizontal and vertical polarization, respectively, and "VV FERM" denotes the computed results using FERM for the vertical-polarization case.

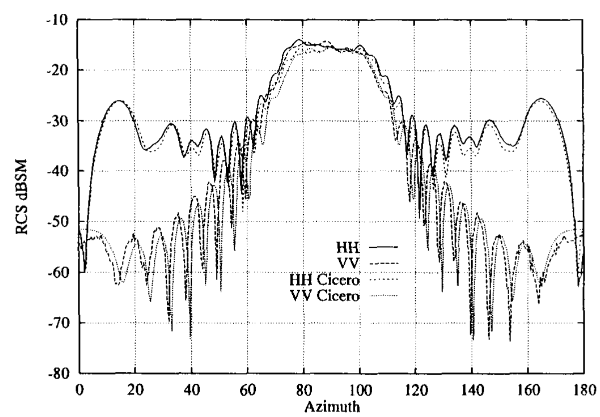

Figure 9. The results for the metallic ogive at 9 GHz. The curves are labeled the same as in Figure 8.

Metallic double ogive

The double ogive consists of two different-size half ogives. One half ogive has a half angle of 46.4 degrees at the tip, a half length of 2.5 inches, and a maximum radius of 1 inch. The other half ogive has a half angle of 22.62 degrees at the tip, a half length of 5 inches, and a maximum radius of 1 inch. Consequently, there is a derivative discontinuity where the two halves join. The double ogive is shown in Figure 10. This target can be analytically expressed as follows:

for -2.5 in $< x <$ 0 in and $-\pi < \psi < \pi$, define

$$g(x) = \left\{ \sqrt{1 - \left(\frac{x}{2.5}\right)^2 \sin^2(46.6°)} - \cos(46.6°) \right\}, \text{ then}$$

$$y = \frac{g(x)\cos\psi}{1 - \cos(46.4°)}$$

$$z = \frac{g(x)\sin\psi}{1 - \cos(46.4°)}$$

for 0 in $< x <$ 5 in and $-\pi < \psi < \pi$, define

$$f(x) = \left\{ \sqrt{1 - \left(\frac{x}{5}\right)^2 \sin^2(22.62°)} - \cos(22.62°) \right\}, \text{ then}$$

$$y = \frac{f(x)\cos\psi}{1 - \cos(22.62°)}$$

$$z = \frac{f(x)\sin\psi}{1 - \cos(22.62°)}$$

In Figures 11 and 12, the RCS characteristics for both horizontal and vertical polarizations are plotted as a function of the azimuthal angle. The elevation angle is zero. Zero degrees azimuth corresponds to the 22.62° tip. The RCS plots in Figure 11 correspond to a frequency of 1.57 GHz, at which the double ogive is approximately one wavelength long. The frequency is 9 GHz in Figure 12.

We note that the VV and HH RCS levels should be equal at both 0° and 180° azimuth.

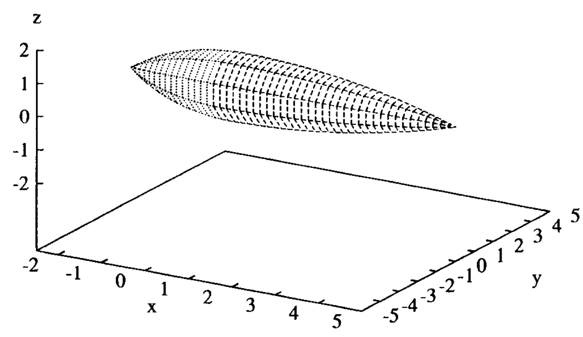

Figure 10. The double ogive.

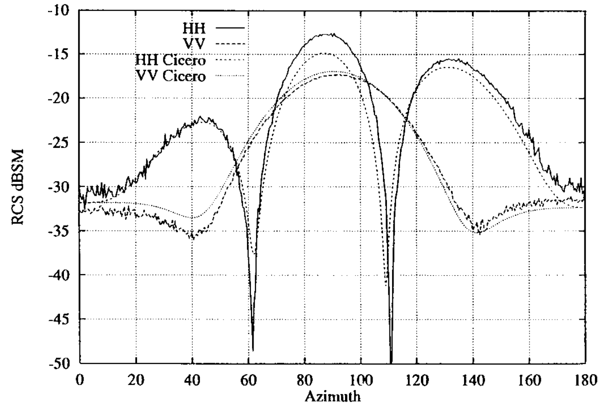

Figure 11. The results for the double ogive at 1.57 GHz. The curves are labeled the same as in Figure 8.

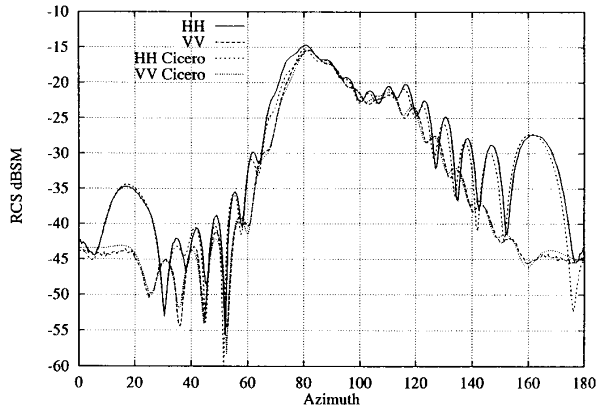

Figure 12. The results for the double ogive at 9 GHz. The curves are labeled the same as in Figure 8.

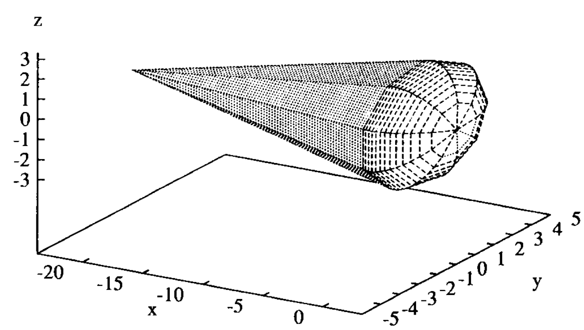

Figure 13. The metallic cone-sphere.

Metallic cone-sphere

The cone-sphere is also a common high-frequency RCS target. The cone-sphere has a half angle of 7 degrees, and a radius of 2.947 inches. The length of the cone part is 23.821 inches, and the side of the cone is tangent to the sphere, to provide a smooth transition and minimum diffraction at the joint. The cone-sphere is illustrated in Figure 13, and the analytical description of the surface is as follows:

for -23.821 in $< x <$ 0 in and $-\pi < \psi < \pi$,

$$y = 0.87145(x + 23.821)\cos\psi$$

$$z = 0.87145(x + 23.821)\sin\psi;$$

for 0 in $< x <$ 3.306 in and $-\pi < \psi < \pi$,

$$y = 2.947\sqrt{1 - \left(\frac{x - 0.359}{2.947}\right)^2}\cos\psi$$

$$z = 2.947\sqrt{1 - \left(\frac{x - 0.359}{2.947}\right)^2}\sin\psi$$

In Figures 14, 15, and 16, the RCS values for both horizontal and vertical polarization are plotted in dBSM, as a function of the azimuthal angle, for two frequencies (869 MHz and 9 GHz). The elevation angle is zero, and the cone-sphere is positioned as in Figure 13. Zero degrees corresponds to incidence normal to the tip. The frequency of 869 MHz was chosen because the cone-sphere is approximately 2 wavelengths long.

There is a large discrepancy between the CICERO calculations and the measurements for azimuth angles near incidence normal to the tip. The cone-spheres will be re-measured this year to resolve this difference.

Metallic cone-sphere with gap

The "metallic cone-sphere with gap" target is the same as the cone-sphere target, except for a gap next to the cone-sphere joint. The gap is ¼ inch wide and ¼ inch deep. The cone-sphere with gap target is illustrated in Figure 17. This target differs from the previous cone-sphere only from 0 in $< x <$ 0.25 in, where it is defined by

for $-\pi < \psi < \pi$,

$$y = 2.697\cos\psi$$

$$z = 2.697\sin\psi$$

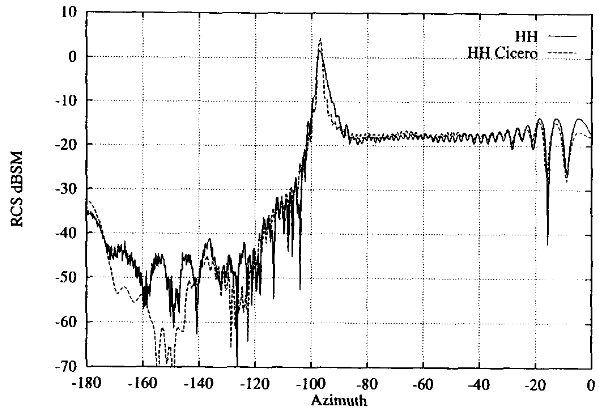

Figure 15. The results for the metallic cone-sphere at 9 GHz, for horizontal polarization. The curves are labeled according the system described for Figure 8.

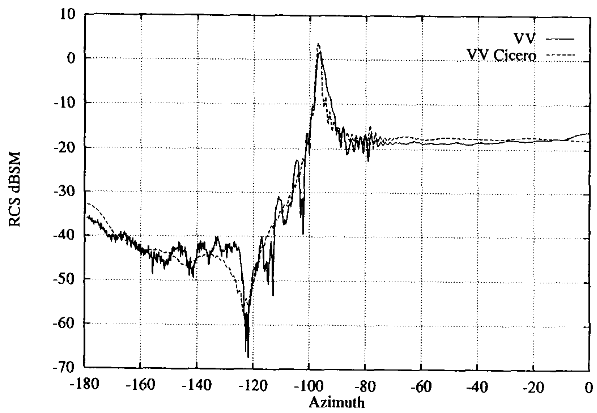

Figure 16. The results for the metallic cone-sphere at 9 GHz, for vertical polarization. The curves are labeled according the system described for Figure 8.

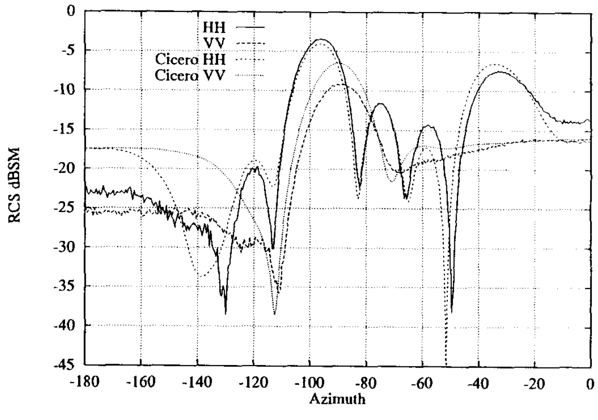

Figure 14. The results for the metallic cone-sphere at 869 MHz. The curves are labeled the same as in Figure 8.

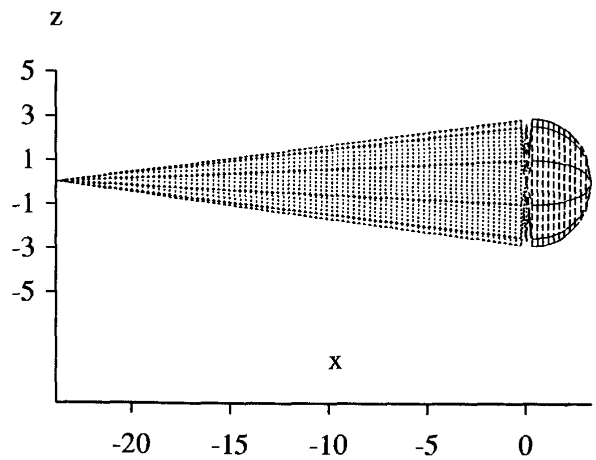

Figure 17. The metallic cone-sphere with a ¼ in gap.

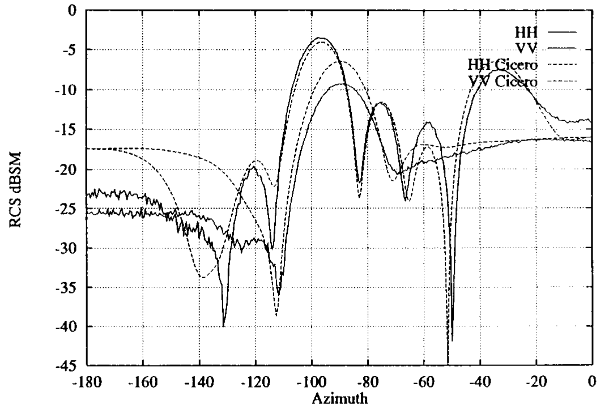

Figure 18. The results for the metallic cone-sphere with a gap at 869 MHz. The curves are labeled the same as in Figure 8.

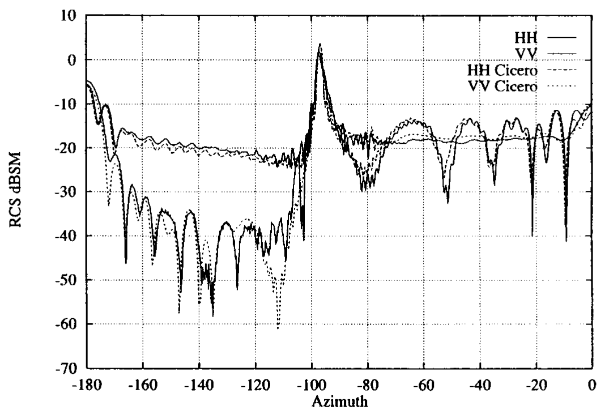

Figure 19. The results for the metallic cone-sphere with a gap at 9 GHz. The curves are labeled the same as in Figure 8.

In Figure 18, the RCS values of both horizontal and vertical polarizations are plotted against azimuthal angle. The elevation angle is zero. Zero degrees azimuth is toward the pointed end. At 869 MHz, this target is approximately two wavelengths in length. At this frequency, the gap is approximately 1/50th of a wavelength, and is hardly visible. Measurement data exist from 700 MHz to 18 GHz in this case. Figure 19 shows results at 9 GHz, where the effect of the gap is pronounced.

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